

In-Situ Monitoring of TID-Induced Timing Degradation in a 16-nm FinFET FPGA-Based RISC-V SoC

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Abstract—We report in-situ monitoring of radiation-induced timing degradation in a 16-nm FinFET UltraScale+ FPGA implementing a RISC-V SoC. X-ray irradiation up to 3 Mrad with 98 sensors revealed temporary delay shifts in UART and system bus.

Index Terms—embedded reliability, on-chip sensors, online delay monitoring, radiation effects, slack-band sensing, X-ray dosimetry.

I. INTRODUCTION

DIGITAL integrated circuits used in radiation-exposed systems are subject to total ionizing dose (TID) effects that can modify transistor parameters, increase leakage, and reduce timing margins in programmable logic fabrics and embedded datapaths [1]–[3]. In parallel, aging-monitoring research has been consolidated and established configurable on-line sensors, representative critical-path selection strategies, and in-depth studies comparing the combined effects of TID, SEU and electromagnetic interference (EMI) in integrated circuits [4]–[7]. However, irradiation studies and path-based aging-monitoring studies are usually reported separately, which motivates in-situ experiments capable of directly observing TID-driven timing shifts in an operating FPGA-based system.

In this work, a Croc RISC-V SoC is implemented on a 16-nm FinFET Artix UltraScale+ FPGA and instrumented with 98 aging sensors distributed across architecturally relevant timing-critical regions. The device is irradiated with X-rays up to approximately 3 Mrad(SiO₂) while sensor alarms, supply current, and temperature are monitored throughout the campaign. The results show detectable alarm changes in the UART TX and OBI demultiplexer regions after 2 Mrad, a temporary recovery after the weekend pause, and reappearance

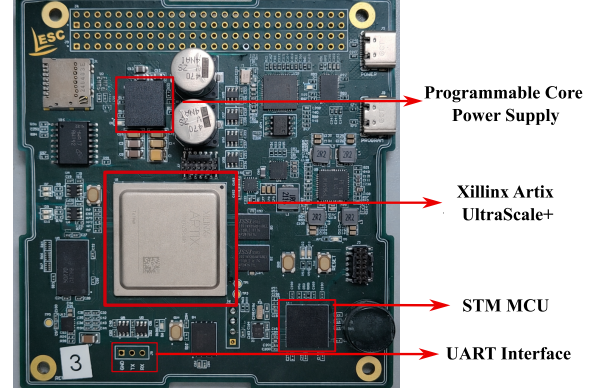


Fig. 1. Annotated top view of the custom FPGA board used in the irradiation campaign. The most relevant board-level features are the programmable FPGA core power supply, FPGA core-current monitoring, and FPGA core-temperature monitoring during operation. The STM MCU is responsible for interfacing and managing the board-level communication and control functions.

of the same alarms by the end of irradiation day 3. Within the 0.694 ns sensor resolution, the study shows that embedded slack-band monitoring can reveal subtle, region-dependent TID-induced timing degradation in a running FPGA-based SoC.

II. EXPERIMENTAL METHODOLOGY

A. FPGA Platform and Board-Level Features

The experiment was implemented on a Xilinx Artix UltraScale+ FPGA fabricated in a 16 nm FinFET technology node (part number *XCAU25P-1FFVB676I*). The device was mounted on a custom CubeSat-form-factor board (10 × 10 cm) developed at the Universidade Federal do Ceará, shown in Fig. 1.

B. Monitored SoC Architecture

The monitored architecture consists of inserting the aging sensor proposed in [8], [9] into timing-critical regions of the Croc SoC [10]. In this work, the evaluated SoC configuration preserved the original Croc architectural organization except for sensor insertion, and the system clock was operated at 20 MHz. The processor subsystem is based on the CVE2 core, an Ibex-derived two-stage pipeline, Harvard-style 32-bit RISC-V processor for embedded control applications [11].

C. Aging Sensor Operation and Calibration

Each aging sensor is connected in parallel to the monitored signal path and uses two clock inputs: the nominal system

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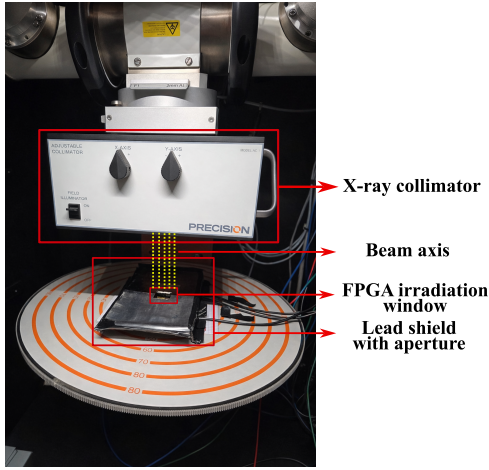


Fig. 3. X-ray irradiation setup used during the experimental campaign. The DUT was placed under a collimated X-ray beam inside the cabinet irradiator, while a lead shield with an aperture localized the exposure to the FPGA region and the board remained powered during irradiation.

Legend: ■ UART TX ■ OBI Demux ■ Fading					[SB2] [SB1]	
Ⓐ Day 1 - Fresh FPGA						
Ⓑ Day 2 - End at 2MRad			Timing degradation	Timing degradation		
Ⓒ Day 3 - Start at 2MRad			Timing recovery	Timing recovery		
Ⓓ Day 3 - End at ~3MRad			Timing degradation	Timing degradation		
Slack Interval	-50ns	(...)	-45.1ns	-44.4ns	-43.8ns	-43.1ns (...)

Fig. 4. The figure highlights the monitored slack bands and the evolution of alarm activation in the UART TX and OBI demux regions. Sensor activation maps at the key stages of the irradiation campaign: (A) fresh FPGA at the beginning of irradiation day 1 (0 Mrad); (B) end of irradiation day 2 (nominally 2 Mrad); (C) start of irradiation day 3 after the weekend pause (nominally 2 Mrad); and (D) end of irradiation day 3 (nominally ~3 Mrad).

active, as shown in Fig. 4(A). In the next interval, *SB2*, the number of active alarms increased to eight in UART TX and two in OBI demux. This pattern was adopted as the baseline sensor state.

As shown in Fig. 5(A), the FPGA temperature started at approximately 40 °C and increased approximately linearly up to about 96 °C by the end of irradiation day 1, when the accumulated dose reached 1 Mrad. Over the same interval, the supply current remained comparatively stable, as indicated in Fig. 5(B), with an average value close to 0.47 A and only a slight upward trend as TID increased. No change in sensor activation was observed at the end of irradiation day 1. Therefore, up to 1 Mrad, no timing shift was detected within the monitored slack bands.

Irradiation day 2 (1 Mrad to 2 Mrad): Between the end of irradiation day 1 and the beginning of irradiation day 2, the FPGA temperature stabilized at approximately 75 °C. At the start of irradiation day 2, with the accumulated dose at 1 Mrad, the sensor state remained unchanged with respect to the end of irradiation day 1.

During irradiation day 2, the temperature increased from about 75 °C to approximately 130 °C as the accumulated dose progressed from 1 Mrad to 2 Mrad, as shown in Fig. 5(C). In parallel, the supply current exhibited an increasing trend with TID, as indicated in Fig. 5(D), with an average value close to

0.9 A and peaks reaching approximately 1.6 A.

At the end of irradiation day 2, after the accumulated dose reached 2 Mrad, a change in the sensor activation pattern was observed. In *SB1*, two additional UART TX sensors became active. In *SB2*, one additional OBI demux sensor became active, as indicated in Fig. 4(B). This result indicates that timing degradation became detectable in the monitored regions.

Irradiation day 3 (restart at 2 Mrad after the weekend pause): After irradiation day 2, the FPGA remained over the weekend at a stabilized temperature of approximately 95 °C. At the beginning of irradiation day 3, with the nominal accumulated dose still at 2 Mrad, the additional sensors that had become active at the end of irradiation day 2 were no longer active, as shown in Fig. 4(C). In other words, the sensor map returned to the same activation pattern observed before the changes detected at the end of irradiation day 2.

During irradiation day 3, the temperature again increased with TID, reaching approximately 150 °C by the end of the irradiation interval, as shown in Fig. 5(E). The current followed the same general trend observed previously, increasing over a significant portion of the day as irradiation progressed, as indicated in Fig. 5(F).

By the end of irradiation day 3, at a nominal accumulated dose of approximately 3 Mrad, the same additional alarms observed at the end of irradiation day 2 reappeared. Specifically, the two additional UART TX sensors in *SB1* became active again, and the additional OBI demux sensor in *SB2* also reappeared, as shown in Fig. 4(D).

IV. DISCUSSIONS

Overall, measurable TID-induced timing degradation became detectable only after 2 Mrad, and only in the UART TX and OBI demux groups. The temporary disappearance of the additional alarms after the weekend pause is consistent with partial annealing reported for irradiated FPGAs and FinFET devices [13]–[15], whereas their reappearance by the end of irradiation day 3 indicates that the underlying degradation persisted under renewed exposure. While the inferred delay shift remained below the sensor resolution (0.694 ns), the proposed sensor infrastructure was still able to detect timing degradation through changes in the alarm activation pattern. This absolute resolution can be further improved in future implementations by increasing the system clock frequency, which reduces the sensor phase step in time and also decreases the nominal path slack. The combined increase in current and temperature is physically consistent with known TID-driven charge trapping, leakage increase, and parametric degradation in MOS and FinFET devices [1], [3], [16]. The observed alarm evolution is compatible with subtle radiation-induced delay degradation, potentially reinforced by increased self-heating during irradiation [14], [17].

V. CONCLUSION

This work demonstrated in-situ timing monitoring of the Croc RISC-V SoC instrumented with 98 aging sensors and implemented on a 16 nm FinFET Artix UltraScale+ FPGA during X-ray irradiation up to approximately 3 Mrad(SiO₂).

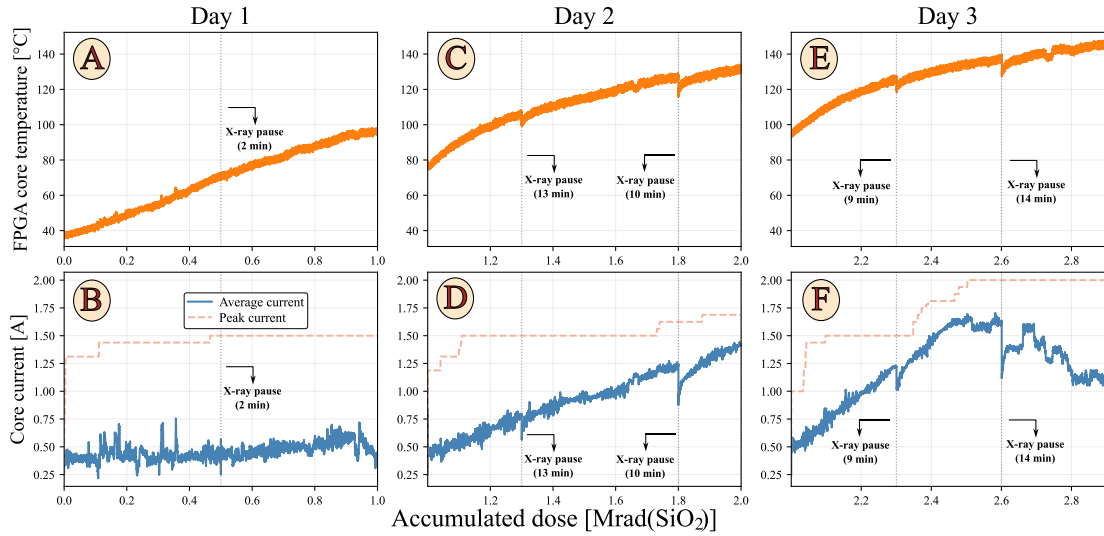


Fig. 5. Supply current and FPGA in-chip temperature as functions of accumulated TID: (A), (C) and (E): temperature during irradiation days 1, 2 and 3, respectively. (B), (D) and (F): current during irradiation days 1, 2 and 3, respectively. Abrupt downward excursions correspond to short interruptions of the X-ray machine.

No alarm change was observed up to 1 Mrad. After 2 Mrad, additional alarms appeared in the UART TX and system bus (OBI demux) groups, disappeared after the weekend pause, and reappeared at the end of irradiation day 3, indicating subtle TID-induced timing degradation with partial recovery behavior.

The inferred delay shift remained under the sensor resolution (0.694 ns), which highlights consistent path time slack erosion and the need to monitor on-line such degradation as function of time. The clock system was set up to 20 MHz to allow precise discrimination of path time slack degradation as function of time, and the capability of the sensors to monitor such degradation. In future work, system clock frequency and sensor's sensitivity will be increased in order to demonstrate that the sensors are not only able to track signal propagation delay increase, but also be used in the capacity of predicting circuit erroneous logic states to be stored in flip-flops before they can actually happen (i.e., allowing system failure prediction).

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